ABDULRHMAN HANY

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EDUCATION

Nile University *Fall 2025 - Present*
Bachelor of Science in Cyber Security

TECHNICAL SKILLS

- **Focus:** Cybersecurity
- **Languages:** Python, C++, C, JavaScript, SQL | Arabic (Native), English (C1 Fluent)
- **Frameworks & Libraries:** React, FastAPI, HTML, CSS
- **Backend & Databases:** Firebase, Supabase, Socket Programming
- **Systems & Tools:** Linux(Kali Linux), Virtual Machines (VMs), Git/GitHub, ESP32 Microcontroller Architecture

SOFTWARE ENGINEERING & SECURITY PROJECTS

Cyber Threat Intelligence & IoT Security

- **Cryptotool CLI:** Engineered an interactive, user-friendly terminal utility optimized for Kali Linux environments to handle real-time data translation. Built functions executing symmetrical ciphers (ROT13) along with Base64, Hexadecimal, and Binary translation streams inside an infinite loop runtime environment.
- **Global Cyber Threat Intelligence Dashboard (Cyber-Attacks-Map / Threat-Pulse):** Developed a real-time, simulated global cyberattack monitoring dashboard using HTML and JavaScript designed for interactive threat intelligence visualization and education.
- **ESP32 Cyber Radar & BLE Signal Tracker:** Built an advanced, dual-mode RF tracking system for the ESP32 architecture to intercept Bluetooth Low Energy (BLE) advertisements and analyze proximity signals.

Security & Systems Tooling

- **Directory Security Scanner:** Engineered a C++ application utilizing the Win32 API to scan and analyze file metadata, specifically optimized to detect "Hidden Executables" as indicators of compromise.
- **Process Manager:** Built a lightweight Windows system utility in C++ interfacing with the Win32 API to capture real-time process snapshots, displaying active PIDs and executable states.
- **Network Port Scanner:** Developed a custom Python-based network reconnaissance utility using socket programming to execute TCP three-way handshakes and map open ports.

Full-Stack & Automation Applications

- **MindSpace (Full-Stack Developer):** Developed a comprehensive full-stack web application and AI compain leveraging a React frontend and a FastAPI backend, integrating Supabase/Firebase for secure user state and persistent data storage.
- **Automated Crypto/Stock Price Notifier:** Programmed an automated Python background utility interfacing with the Binance API to poll real-time crypto fluctuations and dispatch instant alerts.
- **Camera Movement Detection:** Utilized Python and computer vision libraries to capture live video feeds and trigger warning alarms upon detecting pixel movement changes.

Algorithms & Utility Tools

- **Exam Portal Architecture:** Designed the architecture for an academic exam management portal written in Python to handle university assessments, course examinations, and grading.

- **Data Structures Implementation:** Hand-coded foundational computer science data structures in C++, including Singly Linked Lists (insertion, deletion, traversal) and complex Graph Representations with efficient memory management.
- **3D Rendering Script:** Implemented a mathematical "donut" (torus) terminal renderer in Python showcasing 3D projection algorithms and matrix math.
- **Utility & Game Scripts:** Developed an interactive CLI Alarm Clock (C++), an Advanced Calculator (Python) handling 2nd-degree equations/binary parsing, and a terminal Number Guessing Game with dynamic difficulty scaling.

CERTIFICATIONS & LEADERSHIP

- **What Is Generative AI?** – LinkedIn Learning (Instructor: Pinar Seyhan Demirdag)
- **Introduction to AI** – Google on Coursera (Credential ID: 9SMHUB1T3EKH)
- **Scientific Computing with Python** – freeCodeCamp (Credential ID: fcc26c47332-4eac-4814-a15b-7a84bb6ff999-scwp)
- **Mint Ambassadors Project Certificate** – iCareer (Successfully completed professional student leadership and development project engagement)